

Unit 3 TEST Version 3

Total: 48.5/28
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1. What is true about the function, $f(x) = \frac{(x+3)(x^3+4x^2+5x+6)}{2x+7}$ has... Circle your answer.

a. A horizontal asymptote at $y = 0$ b. a horizontal asymptote at $y = \frac{1}{2}$

(c) An oblique asymptote

d. No horizontal/oblique asymptote

2. The function $g(x) = \frac{x-2}{2x+3}$ has a x-intercept of:

(1) a. $\frac{1}{2}$ b. $-\frac{2}{3}$ c. $-\frac{3}{2}$

(d) 2

3. Sketch the graph of $f(x) = \frac{2x+2}{x^3+5x^2+8x+4}$ labelling all key features. Please fill in the table.

6.5 $\frac{2x+2}{x^3+5x^2+8x+4}$

$f(-1) = 0$

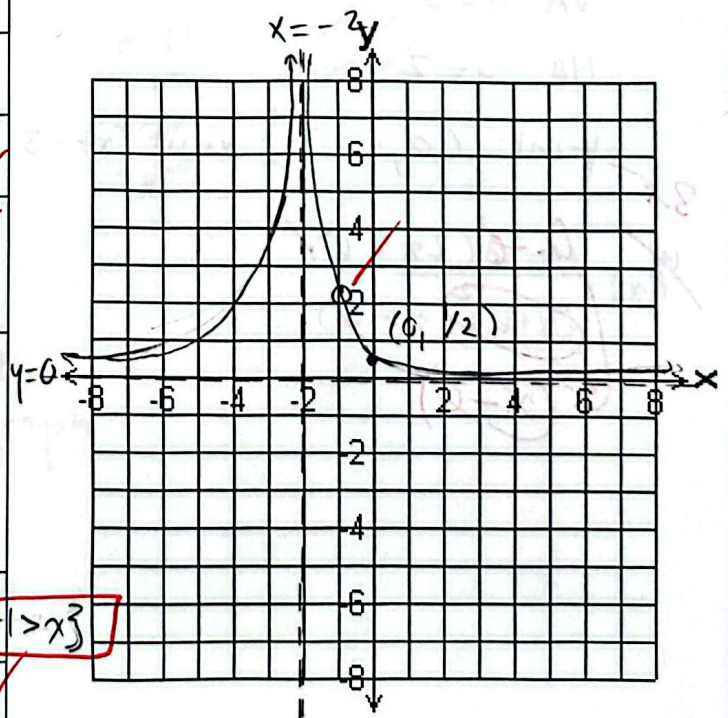
$$\begin{array}{r|rrrr} -1 & 1 & 5 & 8 & 4 \\ + & & -1 & -4 & -4 \\ \hline x & 1 & 4 & 4 & 0 \end{array}$$

$$\frac{2(x+1)}{(x+1)(x+2)^2}$$

hole @ $x = -1$

$$= \frac{2}{(x+2)^2}$$

Vertical Asymptote(s)	$x = -2$
Horizontal Asymptote	$y = 0$
x-intercept(s)	N/A
y-intercept (exact value)	$(0, \frac{1}{2})$
End Behaviour	$x \rightarrow +\infty, f(x) \rightarrow 0^+$ $x \rightarrow -\infty, f(x) \rightarrow 0^+$
Behaviour near Vertical Asymptote(s)	$x \rightarrow -2^+, f(x) \rightarrow +\infty$ $x \rightarrow -2^-, f(x) \rightarrow +\infty$ $x \rightarrow _, f(x) \rightarrow _$ $x \rightarrow _, f(x) \rightarrow _$
Domain	$\{x \in \mathbb{R} \mid x \neq -2, x < -1, -1 > x\}$
Range	$\{y \in \mathbb{R} \mid y \neq 0\}$



7.5

4. Solve the inequality $\frac{x^2-10x-12}{x^2-2x-8} > 3$ using an interval table.

(4) $\frac{x^2-10x-12}{x^2-2x-8} - \frac{3}{1} > 0$

$$\frac{x^2-10x-12-3(x^2-2x-8)}{(x-4)(x+2)} > 0$$

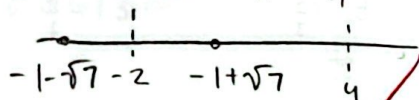
$$\frac{x^2-10x-12-3x^2+6x+24}{(x-4)(x+2)} > 0$$

$$\frac{-2x^2-4x+12}{(x-4)(x+2)} > 0$$

$$\frac{-2(x^2+2x-6)}{(x-4)(x+2)} > 0$$

$$x\text{-int: } -1 \pm \sqrt{7}$$

$$VA: x = 4, -2$$



-3.65
1.65

Interval	$(-\infty, -1-\sqrt{7})$	$(-1-\sqrt{7}, -2)$	$(-2, -1+\sqrt{7})$	$(-1+\sqrt{7}, 4)$	$(4, \infty)$
	(-)	(+)	(-)	(+)	(-)

Solution: $f(x) > 0$ when $(-1-\sqrt{7}, -2) \cup (-1+\sqrt{7}, 4)$ $x \neq 4, -2$

5. Determine a possible equation for the following function.

hole @ $x=6$

linear / 3th

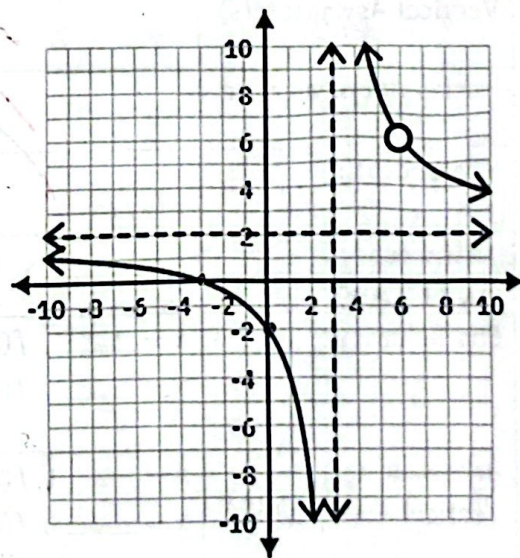
VA $x=3$

HA $y=2$

y-int: $(0, -2)$ x-int: $x=-3$

3.5

$$f(x) = \frac{(x+6)^2(2x+6)}{(x+6)^2(x-3)(x-6)}$$



2.5

6) Determine the key features to help you sketch $f(x) = \frac{x^3 + 4x^2 - 9x - 36}{x^2 + 4x + 3}$.
Label all key features on the graph.

$$x^3 + 4x^2 - 9x - 36$$

$$(x+1)(x+3)$$

$$f(3) = 0 \rightarrow (x-3)$$

$$\begin{array}{r|rrrr} 3 & 1 & 4 & -9 & -36 \\ + & & 3 & 21 & 36 \\ \hline x & 1 & 7 & 12 & 0 \end{array}$$

$$(x+5)(x+4)(x-3)$$

$$(x+1)(x+3)$$

hole @ $x = -3$

$$VA: x = -1$$

$$x\text{-int: } x = 3, -4 \quad (3, 0) \quad (-4, 0)$$

$$y\text{-int: } (0, -12)$$

$$\begin{array}{r} 5.5 \\ x^3 + 4x^2 + 3x \overline{) x^3 + 4x^2 - 9x - 36} \\ \underline{-x^3 + 4x^2 + 3x} \\ -12x - 36 \end{array}$$

$$\underline{-x^3 + 4x^2 + 3x} \downarrow$$

$$-12x = 36$$

$$-12x = 36$$

$$\underline{-12} \quad \underline{-12}$$

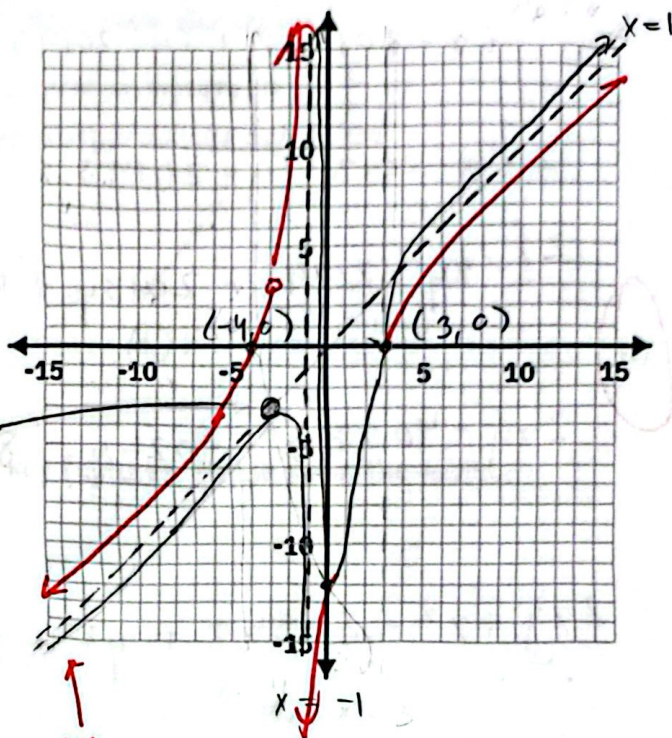
$$OA: x$$

which one?

$$x = -3'$$

where it crosses graph

$$\begin{array}{r|rr} x+1 & x^2 + x - 12 \\ - & x^2 - x \\ \hline & 2x - 12 \\ & -2x + 2 \\ \hline & -10 \end{array}$$



$$\begin{array}{cc} x \rightarrow -1^+ & -\infty \\ x \rightarrow -1^- & +\infty \\ x \rightarrow +\infty & - \\ x \rightarrow -\infty & - \end{array}$$

5.5

7. If $\frac{(x-2013)(y-2014)}{(x-2013)^2 + (y-2014)^2} = -\frac{1}{2}$, determine the value of $x+y$.

$$\frac{(x-2013)(y-2014)}{(x-2013)^2 + (y-2014)^2} = -\frac{1}{2}$$

$$2(x-2013)(y-2014) = -[(x-2013)^2 + (y-2014)^2]$$

$$2[xy - 2014x - 2013y + 4054182] = -(x^2 - 4026x + 4052169 + y^2 + 4028y + 4056196)$$

$$2xy - 4028x - 4026y - 8108369 = -x^2 + 4026x - 8108365 - y^2 - 4028y$$

$$2xy + x^2 - 8054x + y^2 + 2y + 1 = 0$$

$$x(x-8054) + y(y+2) + 2xy + 1 = 0$$

$$x = 8054 \quad y = -2$$

$$8054 + (-2) = 8052$$

let a rep. $(x-2013)$ let b rep. $(y-2014)$

$$\frac{ab}{a^2 + b^2} = -\frac{1}{2}$$

$$2[ab] = -[a^2 + b^2]$$

$$2ab = -a^2 - b^2$$

$$a^2 + 2ab + b^2 = 0$$

$$(a+b)^2 = 0$$

$$[(x-2013) + (y-2014)]^2 = 0$$

$$(x+y-4027)^2 = 0$$

$$x^2 + y^2 + 16216729 = 0$$

$$x = y + 4027i$$

$$\boxed{x+y = 4027}$$

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